Fourth Normal Form

4FN

Understanding Multivalued Dependencies and Database Normalization

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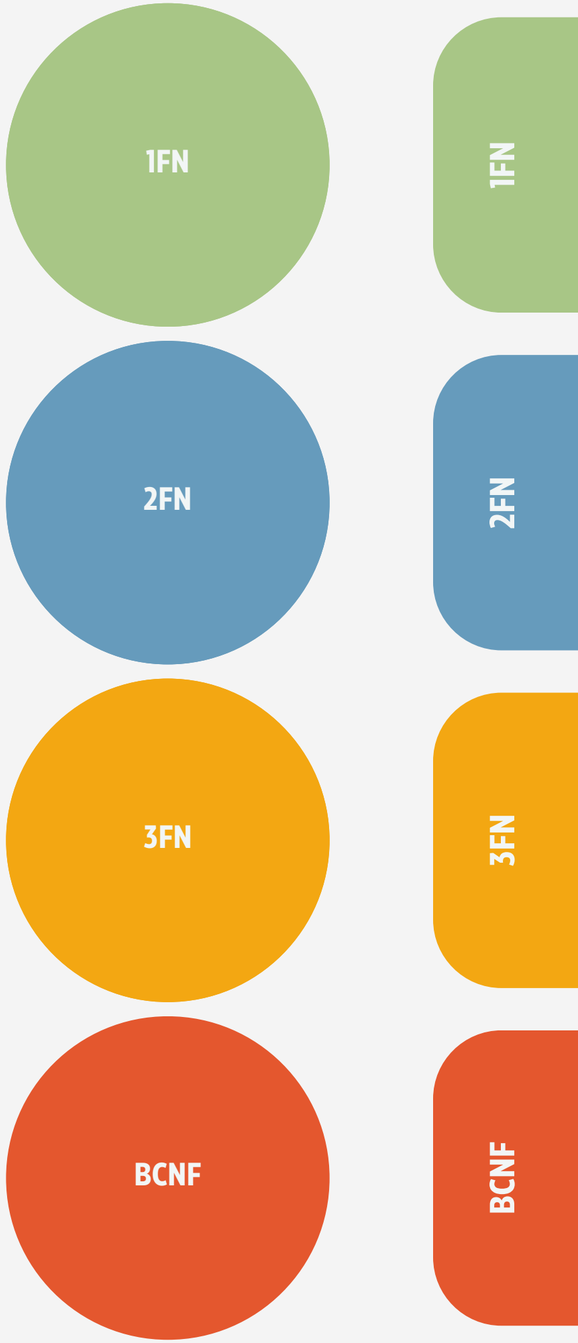

The Origins of 4NF

- Defined by computer scientist Ronald Fagin in 1977.
- Built to extend the Boyce-Codd Normal Form (BCNF).
- The Goal: Solve a specific flaw in relational models called Multivalued Dependencies.

RONALD FAGIN

1FN

2FN

3FN

BCNF

Multivalued Dependency (MVD)

- One attribute determines multiple values of a second attribute...
- ...AND multiple values of a third attribute.
- Crucial condition: The second and third attributes are completely independent of each other.

Notation: A ->-> B

EXAMPLE

Developers

Developer	Project	Mastered_Language
Carlos	Project Alpha	Python
Carlos	Project Alpha	Java
Carlos	Project Beta	Python
Carlos	Project Beta	Java

What is Fourth Normal Form?

A table is in 4NF if and only if:

1. It is already in BCNF (Boyce-Codd Normal Form).
2. It has NO non-trivial multivalued dependencies.

IMPORTANT "No table should contain two or more independent, multivalued facts about the same element."

EXAMPLE

Developers

Developer	Project	Mastered_Language
Carlos	Project Alpha	Python
Carlos	Project Alpha	Java
Carlos	Project Beta	Python
Carlos	Project Beta	Java

WHAT IF CARLOS LEARNS 'C++'?

WE ARE FORCED TO ADD TWO NEW ROWS (ONE FOR EACH PROJECT)

PROBLEM = REDUNDANCY

The 4NF Solution

Developer_Project

Developer	Project
Carlos	Project Alpha
Carlos	Project Beta

Developer_Language

Developer	Mastered_Language
Carlos	Python
Carlos	Java

PROBLEM SOLVED

Add only one row when Carlos learns 'C++'

The Hamburger Stand Problem

- Scenario: A food stand called Sin Limites Burger offers 40 types of burgers (Turkey, Chorizo, etc.).
- The Delivery: The stand also delivers to 3 cities (New York, Chicago, Miami).
- The Conflict: Burger types and delivery zones have no relationship to each other. They only depend on the stand.

Stand_Name	Burger_Type	Delivery_City
Sin Limites Burger	Turkey Burger	New York
Sin Limites Burger	Turkey Burger	Chicago
Sin Limites Burger	Turkey Burger	Miami
Sin Limites Burger	Chorizo Burger	New York
Sin Limites Burger	Chorizo Burger	Chicago
Sin Limites Burger	Chorizo Burger	Miami

1FN

2FN

3FN

BCNF

The Math of Redundancy

Why 4NF is mathematically necessary??

Without 4NF (Unnormalized)

- 40 Burgers × 3 Delivery Zones = 120 rows!
- Result: 120 lines of highly repetitive and redundant data for just one stand.

Solution

Stand_Burgers

Stand_Name	Burger_Type
Sin Limites Burger	Turkey Burger
Sin Limites Burger	Chorizo Burger
... (up to 40)	...

Stand_Delivery

Stand_Name	Delivery_City
Sin Limites Burger	New York
Sin Limites Burger	Chicago
Sin Limites Burger	Miami

With 4NF (Normalized)

- Table 1 (Stand + Burgers): 40 rows
- Table 2 (Stand + Delivery): 3 rows
- Result: 40 + 3 = 43 rows in total.

Stand_Burgers

Stand_Name	Burger_Type
Sin Limites Burger	Turkey Burger
Sin Limites Burger	Chorizo Burger
... (up to 40)	...

Stand_Delivery

Stand_Name	Delivery_City
Sin Limites Burger	New York
Sin Limites Burger	Chicago
Sin Limites Burger	Miami

Functional Dependency

Stand_Name	Owner
Sin Limites Burger	Jonny
Sin Limites Burger	Jonny
Sin Limites Burger	Jonny

1FN

2FN

3FN

BCNF

EXCERCISE

Context: A university course (CS101) has multiple approved Instructors and multiple required Textbooks.

Crucial Fact: Any approved instructor can teach using any of the approved textbooks. They are completely independent.

- **If the department approves a new textbook ("Algorithms 101"), how many rows must be added?**
- **If Prof. Davis retires, how many rows must be deleted?**
- **How do we normalize this to 4NF?**

Course	Instructor	Textbook
CS101	Prof. Smith	Intro to C
CS101	Prof. Smith	Data Structures
CS101	Prof. Johnson	Intro to C
CS101	Prof. Johnson	Data Structures
CS101	Prof. Davis	Intro to C
CS101	Prof. Davis	Data Structures

PROBLEM SOLUTION

1. Add 3 rows (one for each existing instructor). Insertion Anomaly!
2. Delete 2 rows (Prof. Davis with each textbook). Deletion Anomaly!

Course_Instructor

Course	Instructor
CS101	Prof. Smith
CS101	Prof. Johnson
CS101	Prof. Davis

Course_Textbook

Course	Textbook
CS101	Intro to C
CS101	Data Structures

1FN

2FN

3FN

By splitting the table, adding a new textbook requires adding only ONE row in the Course_Textbook table.

Thanks for your attention

<3

1FN

2FN

3FN

BCNF